

Delhi Technological University

Department of Computer Science & Engineering

Subject: Object-Oriented Programming Lab

Subject Code: CS203

Programme: B. Tech (CSE)-3rd Semester

LIST OF PROGRAMS

1. Introduction to Object-Oriented Programming. Discuss features of Object-Oriented Programming.
2. Write a program to print "Hello World" in C++.
3. Write a program to perform the addition of two numbers in C++.
4. Perform the sum of natural numbers in C++.
5. Implement any type of sorting of N numbers.
6. Write a program for the multiplication of two matrices in C++.
7. Create a class Student in C++ with data members rollNo and name. Include member functions getDetails() to input values and displayDetails() to display them. Create an object of the class in main() to demonstrate the use of a class and an object in C++.
8. Define a class BankAccount with data members accountNumber and accountHolder. Use a default constructor to display default values assigned, then use a parameterized constructor to initialize the member with some values and a destructor to display a message when the account object is destroyed. Include a function to display the account details. Create multiple objects in main() to demonstrate constructor and destructor usage.
9. Create a class Student with private data members name, roll_no, and marks. Use a parameterized constructor to initialize them, and a copy constructor to copy the details of another student object. Include a member function to display the details.
10. Create a class Student, which has data members as name, branch, roll no., age, sex, and five subjects. Display the name of student & his percentage who has more than 70%. Use array of objects. [Array of objects]
11. Perform the addition of two complex numbers using constructor overloading. The first constructor, which takes no arguments, is used to create objects that are not initialized. The second, which takes one argument, is used to initialize real and imaginary parts to equal values, and the third, which takes two arguments, is used to initialize real and imaginary parts to two different values.
12. Create a base class basicinfo with data members name, roll no, sex, and two member functions getdata and display. Derive a class PhysicalFit from basic_info, which has data members height and weight, and member functions getdata and display. Display all the information using an object of a derived class. (Single Inheritance)
13. Create a class first with data members book no, book name, and member functions getdata and putdata. Create a second class with data members author name, publisher,

and members getdata and showdata. Derive a class third from first and second with data members no of pages and year of publication. Display all this information using an array of objects of the third class. (Multiple Inheritance)

14. Design three classes STUDENT, EXAM, and RESULT. The STUDENT class has data members such as roll number and name. Create a class EXAM by inheriting the STUDENT class. The EXAM class adds data members representing the marks scored in six subjects. Derive the RESULT from the EXAM class and has its own data members such as total marks. WAP to model this relationship. (Multilevel Inheritance)
15. Using the concept of function overloading, write a function for calculating the area of a triangle, circle, and rectangle.
16. Write a C++ program to print the area of a rectangle by creating a class named Area having two functions. The first named setDim() takes the length and breadth of the rectangle, and the second named getArea() returns the area of the rectangle.
17. Write a program to find the greatest of two given numbers in two different classes using a friend function.
18. Implement a class string containing the following functions:
 - Overload + operator to carry out the concatenation of strings.
 - Overload = operator to carry out string copy.
 - Overload <= operator to carry out the comparison of strings.
 - Function to display the length of a string.
 - Function tolower() to convert upper case letters to lower case. -
 - Function toupper() to convert lower case letters to upper case.
19. Write a program to overload the unary increment (++) operator
20. Write a program to define the function template for calculating the square of given numbers with different data types.
21. Write a program to read the class object of student info such as name, age, sex, height and weight from the keyboard and to store them on a specified file using read() and write() functions. Again, the same file is opened for reading and displaying the contents of the file on the screen.
22. Create a base class called SHAPE. Use this class to store two double type values. Derive two specific classes called TRIANGLE and RECTANGLE from the base class. Add to the base class, a member function getdata to initialize base class data members and another member function display to compute and display the area of figures. Make a virtual function and redefine this function in the derived classes to suit their requirements. Using these three classes design a program that will accept driven of a TRIANGLE or RECTANGLE interactively and display the area.
23. Write a program to illustrate how to define and declare a class template for reading two data items from the keyboard and to find their sum.
24. Introduction to Java. Illustrate the Compilation and Execution Process in Java.

25. Implement single-level, multi-level and hierarchical inheritance in Java.
26. Write a program to demonstrate the use of the final keyword with data members, functions, and classes.
27. Write a program in Java to implement multiple inheritance in Java through interfaces.
28. Write a program to show the use of the 'Wrapper' Class in Java.
- 29 (a.) Write a program in Java to implement static polymorphism (Method Overloading).
- (b.) Write a program in Java to implement dynamic polymorphism (Method overriding).
30. Show the use of the 'super' keyword in Java
31. Write a program to show the use of the 'Abstract' Class in Java
32. Write a program to show the use of the 'static' keyword in Java.
33. Create a customized exception and make use of all 5 exception keywords.
34. Design a class that performs String operations (Equal, Reverse the string, change case).
35. Design a program to demonstrate multithreading using the Thread Class.
 - (a) Write a program to print a table of 2 numbers without using synchronization using multithreading.
 - (b) Write a program to print a table of 2 numbers using synchronization using multithreading.
36. Design a program to create a basic calculator using Applet and Event Handling.
37. Create a Java program that stores student names using a HashSet to ensure no duplicates are added.
38. Write a Java program using ArrayList to maintain a list of fruits. Allow the user to add fruits to the list and then display them one by one. Take input of 5 student names and display the final unique list.
39. Design a program to read a text file and after printing that on-screen write the content to another text file.
40. Convert the content of a given file into the uppercase content of the same file.
41. Develop a scientific calculator using swings.

PROJECT

Create any game or any other GUI in C++ or Java.